

ROUTING PROTOCOL FAILURE ANALYSIS

A root-cause investigation into the March 2024 network disruption at NimbusEdge

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Compiled from the incident report, network topology diagram, root-cause analysis report, NOC change log, and client SLA breach notice.

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CASE SNAPSHOT

In March 2024, six enterprise clients lost service for roughly nine hours. The post-incident review uncovered four unrelated routing-protocol faults working together: a hop-count ceiling reached in RIP, sluggish convergence in IGRP, a mismatched Area ID in OSPF, and a BGP failover policy that never triggered.

01 | INTRODUCTION

NimbusEdge Technologies suffered a significant network disruption in March 2024, leaving six enterprise clients without service for close to nine hours. Rather than tracing back to a single cause, the investigation found that four independent routing-protocol problems had surfaced across different segments of the network at once: RIP running into its hop-count ceiling, IGRP responding too slowly to a degraded link, OSPF being blocked by a configuration mismatch, and BGP's automatic failover never engaging.

Each failure is examined on its own terms in the sections that follow — what occurred, the underlying cause, why it went unnoticed for as long as it did, and the fix being recommended — before pulling the four cases together into a single set of conclusions about how the network is designed and operated.

02 | INCIDENT OVERVIEW

PROTOCOL	LOCATION	ROOT CAUSE
RIP	Chennai HQ	A route to the delivery department climbed to 16 hops and was dropped as unreach
IGRP	Bengaluru – Pune – Hyderabad	A degraded link stayed in use too long before traffic shifted (slow convergence)
OSPF	Core Backbone – Hyderabad	A wrong Area ID stopped the neighbor adjacency from ever forming
BGP	ISP-A / ISP-B Peering	The automatic switch to the backup ISP failed to fire as intended

03 | RIP — CHENNAI HQ CAMPUS

What happened

Chennai HQ was still running RIP v2 when a new subnet was added for the delivery department. The path to that subnet stretched to 16 hops. Because RIP only treats routes of up to 15 hops as usable, the new subnet was quietly dropped from the routing table. Nobody noticed until clients began filing support tickets — routine monitoring never flagged the problem.

Why 16 hops broke the route

Hop count is the only metric RIP uses to judge a path — essentially, how many routers a packet has to pass through. Anything from 1 to 15 hops counts as reachable; 16 is treated as infinite distance, i.e., unreachable. The moment the delivery subnet's path hit that 16th hop, RIP simply stopped advertising it as usable.

Why the fault slipped past detection

This was a routing-logic failure, not a hardware one — every router and interface kept working normally, so link-level monitoring showed a clean bill of health. The root-cause review pointed to a monitoring blind spot: route reachability simply wasn't watched as closely as physical link status, and that gap combined with RIP's built-in 15-hop limit to hide the fault.

RECOMMENDED FIX

Move Chennai HQ off RIP and onto OSPF. OSPF has no hop-count ceiling and scales considerably further, which suits the site's continued growth. RIP was a sensible original choice for a small, static campus thanks to its simplicity — the trouble began once the site outgrew that original scope.

RULE OF THUMB 1–15 hops = reachable | 16 hops = unreachable

04 | IGRP — BENGALURU–PUNE–HYDERABAD

What happened

A fault on the ISP side degraded the Bengaluru–Pune fibre link, cutting effective throughput to around 2 Mbps and pushing latency up. A better path was available via Pune–Hyderabad–Bengaluru, but IGRP was slow to move traffic onto it, and clients on the affected segment experienced severe application lag in the meantime.

What the IGRP metric shows

Classic IGRP scores each path mainly on bandwidth and delay. Working through the math for a 2,000 kbps link, the bandwidth term alone comes to $10,000,000 \div 2,000 = 5,000$ before delay is factored in and scaling applied. Since delay had also increased, the degraded path's overall score should have worsened even further — and the healthier route should have won out once its full score came in lower.

One caveat: the source material doesn't include complete delay figures for both routes, so a precise side-by-side score comparison would mean assuming numbers that aren't actually in the documentation.

Why this was really a convergence issue

IGRP's approach to scoring wasn't the problem — factoring in both bandwidth and delay is more useful than hop count alone. The real issue was how long it took the network to react: conditions changed, but the shift to the healthier path lagged well behind. In other words, IGRP had the right picture of the network but was slow to act on it.

RECOMMENDED FIX

Move the regional network onto OSPF. IGRP was a genuine step up from RIP in its day, since it accounts for bandwidth and delay rather than just hop count — but a regional network at this scale needs OSPF's faster convergence and stronger scalability, paired with active monitoring of both link performance and how long routing convergence actually takes.

05 | OSPF — CORE BACKBONE MIGRATION

What happened

NimbusEdge was in the process of moving its core network onto multi-area OSPF, with the Hyderabad site meant to sit in Area 3. Instead, the Hyderabad router ended up configured with the wrong Area ID. Since that ID didn't match the design, the router could never form the OSPF adjacency it needed to, and a routing black hole persisted until a later change record corrected the setting.

Why the Area ID matters

Before two OSPF routers can become neighbors, they need to agree on several settings: Area ID, Hello interval, Dead interval, authentication (if used), and matching network types. Here the mismatch was specifically the Area ID — with the two routers disagreeing on which area they belonged to, no neighbor relationship could ever form.

Why it was hard to catch

A physical link failure tends to be obvious, since the interface itself drops. Here, the physical connection stayed up while the routing layer quietly failed underneath it, so basic link monitoring showed nothing wrong. The root-cause review also flagged a gap in change management: the misconfiguration wasn't caught because validation after the change wasn't thorough enough.

How it should be fixed

- Correct the Hyderabad router's Area ID to the intended Area 3 assignment.
- Confirm the OSPF neighbor relationship reaches FULL state.
- Verify route/LSA exchange and inspect the routing table.
- Run an end-to-end connectivity test.
- Introduce peer review, a formal change checklist, and mandatory adjacency verification after every change.

Compared with the older RIP/IGRP setup, OSPF brings faster convergence, better scalability, and much clearer link-state visibility — which makes it the more suitable protocol for a backbone spanning multiple sites.

06 | BGP — ISP PEERING AND FAILOVER

What happened

NimbusEdge ran two ISP links — ISP-A as the primary and ISP-B as backup. When ISP-A went down, BGP was supposed to withdraw the routes through it and pick up ISP-B automatically. That handover never happened, so engineers had to step in manually — a delay that both stretched out the outage and led directly to the SLA breach.

Key BGP attributes

ATTRIBUTE	MEANING
LOCAL_PREF	How strongly an outbound path is preferred within NimbusEdge's own AS — higher wins.
AS-PATH	The chain of autonomous systems a route has crossed; shorter paths are generally favored.
MED	Signals the preferred entry point between two peered autonomous systems; lower is generally preferred.

The root-cause report traces the failed handover specifically to a LOCAL_PREF / AS-PATH policy problem.

Why an unverified BGP policy is a real risk

BGP governs how the network talks to external autonomous systems and ISPs, so a policy error here can affect a large share of outbound traffic. The chain of events played out as: ISP-A failed → the automatic failover didn't trigger → engineers had to step in by hand → recovery was delayed → clients felt the impact → the SLA was breached.

What the policy should look like

Under normal conditions, ISP-A should carry a higher preference than ISP-B — for instance, a LOCAL_PREF of 200 for ISP-A against 100 for ISP-B. The exact numbers are illustrative; what matters is that ISP-A stays primary and ISP-B stays backup. Once ISP-A's route is withdrawn, ISP-B should become the best available path automatically. NimbusEdge should also be running scheduled, controlled failover drills.

BEFORE	AFTER
ISP-A went down, but the policy in place didn't trigger the expected automatic switch to ISP-B.	ISP-A holds a higher LOCAL_PREF as primary; ISP-B holds a lower one as backup. When ISP-A fails, its route is withdrawn and ISP-B takes over automatically.

07 | OVERALL CONCLUSION

The NimbusEdge outage wasn't the result of one isolated fault but of four separate routing failures landing at once: RIP ran into its built-in 15-hop limit, IGRP reacted too slowly to a degraded link, OSPF was blocked by an incorrect Area ID, and BGP failed to deliver the automatic ISP failover it was designed to provide.

The common thread

Reliable routing takes more than picking the right protocol. It also depends on correct configuration, monitoring that actually reflects routing state, disciplined validation after every change, and regular, realistic failure testing.

CHOOSE the right protocol for the network's scale.	CONFIGURE routing parameters correctly.	MONITOR routing state, not just link status.	TEST failover before a real incident forces it.
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IN SIMPLE TERMS

Choose the right protocol for the network's scale, configure it correctly, monitor actual routing state rather than just link status, and test what happens when a link or path fails — before a real incident makes that test for you.

08 | DOCUMENTS REFERENCED

- Incident_Report_NimbusEdge_March2024.pdf
- Network_Topology_Diagram_NimbusEdge.pdf
- RCA_Report_Q1_2024.docx
- NOC_Change_Log_Feb–Mar_2024.xlsx
- Client_SLA_Breach_Notice.pdf